Assignment – 2

Subject Name & Code	Semester	Issue Date	Due Date
Object Oriented Design & Programming / 01AI0103	2nd	26-03-2025	10-03-2025

Sr. No.	Question	CO
1	What is Inheritance? Explain the different types of inheritance.	CO2
2	What is Operator Overloading? Give example of unary operator overloading.	CO2
3	Explain the following Data Type conversion with example: 1. Basic to Class 2. Class to Basic	CO4
4	Write a C++ program to add to complex numbers using Binary Operator overloading.	CO4
5	Write a C++ program to perform the following Inheritance: 1.**Multiple Inheritance** (Create a Player class with an attribute playerName, a Sport class with an attribute sportType, and a Athlete class that inherits Player and Sports Class; add a function in Athlete to display the details.) 2. **Hierarchal Inheritance** (Create a Vehicle class with an attribute brand. Derive two classes, Car and Bike, from Vehicle; add functions in Car and Bike to display their details.)	CO4
6	Differentiate between Run-time and Compile Time polymorphism.	CO2
7	Explain the concept of Pure Virtual functions with example	CO2
8	Explain the following IOS format functions: 1.Width() 3.Precision() 2.Setf() 4.Unsetf()	CO3
9	Write a program that reads data from one file and copy it to other file.	CO3
10	Explain the different file opening modes.	CO3
11	Write Short notes on: 1.File Pointer 3.Pitfalls of Operator Overloading 2.File Input and Output Stream 4.Template	CO3
12	Explain the concept of Function template with example.	CO3
13	Explain the concept of Class template with example.	CO3
14	Write a program to overload function template.	CO3
15	What is Exception handling? Write a program to show the concept of Exception handling using try and catch block.	CO3